SANJEEV KUMAWAT

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.50
RAJASTHAN PUBLIC SR. SEC. SCHOOL, PAOTA	Class XII (BOARD OF SECONDARY EDUCATION, RAJASTHAN)	2022	93.40%
JYOTI PUBLIC SR. SEC. SCHOOL, JAISINGHPURA, VIRATNAGAR	Class X (BOARD OF SECONDARY EDUCATION, RAJASTHAN)	2020	93.50%

RESEARCH

Machine Learning for Photometric Redshift Estimation

(MSc Research Project)

- Investigating **photometric redshift estimation of radio-selected galaxies** using machine-learning and metric-learning techniques.
- Developing and comparing **regression and classification models**, including kNN, Random Forest, LMNN, AdaBoost, and XGBoost.
- Evaluating **model robustness and generalization** through cross-validation, hyperparameter optimization, and multi-dataset validation using standard astronomical performance metrics.

PROJECTS

GPU-Accelerated Physics Based Urban Flood Modelling

(INSPIRE Summer Project 2026)

- Developed a **2D physics-based hydrodynamic flood solver**, using **ANUGA as the reference implementation** for governing-equation formulation and numerical validation.
- Implemented **GPU-accelerated numerical kernels** using **CUDA, JAX/XLA, and CuPy**, targeting parallel execution of computationally intensive solver operations.
- Conducted **CPU-GPU performance benchmarking and numerical-equivalence validation**, evaluating runtime, scalability, and solution consistency.

Damped Harmonic Oscillator with Stochastic Forces

(Computational Physics Course Project)

- Simulated a damped harmonic oscillator under stochastic forces to study noise-driven dynamics and system response.
- Implemented numerical methods to solve differential equations and analyzed behavior under varying damping and noise strengths.

SKILLS

Programming: Python

AI/ML: PyTorch, Scikit-learn, XGBoost

Web Development: HTML, CSS, JavaScript

Tools: Git, GitHub, VS Code

RELEVANT COURSES

Machine Learning for Physical Sciences - Computational Physics - Linear Algebra - Statistical Mechanics

POSITIONS OF RESPONSIBILITY

Technical Core Member - Cepheid, Astronomy & Astrophysics Club, IIT Hyderabad

- Contributing to **"Identifying Exoplanets with Deep Learning"**, applying deep learning to astronomical data and exoplanet detection.
- Coordinated & volunteered for **Cosmic Quest**, an astronomy quiz involving ~50 school students, along with other event activities.

Core Member - Nakshatra, Astronomy & Astrophysics Club, IIT Hyderabad

- Manage and maintain the **club website**, handling technical updates, content management, and improvements.

Volunteer - Extra Mural Lectures (EML), IIT Hyderabad

- Supported the organization and execution of a distinguished-speaker event featuring **NASA-Axiom Mission Pilot Shubhanshu Shukla**.

ACHIEVEMENT

DST INSPIRE Scholar

- Awarded the **Innovation in Science Pursuit for Inspired Research (INSPIRE) Scholarship** by the Department of Science & Technology (DST), Government of India, for academic excellence and continued pursuit of science and research.

HOBBIES

Cricket - Badminton - Football - Running - Volleyball